

Spotted Python
Week 7 Tutorial

- BPT Diagram
- Reading papers
- Finding papers
- Overleaf
- LaTeX Packages

BPT Diagram

- Simple Diagnostic Diagram (Kewley+2006)
  - *HII & Composites:* $\log\left(\frac{[OIII]}{[OII]}\right) < -1.701\log\left(\frac{[OI]}{H_\alpha}\right) - 2.163$
  - *LINERS:* $\log\left(\frac{[OIII]}{[OII]}\right) < 1.0\log\left(\frac{[OI]}{H_\alpha}\right) + 0.7$
  - *Seyferts:* $\log\left(\frac{[OIII]}{[OII]}\right) > 1.0\log\left(\frac{[OI]}{H_\alpha}\right) + 0.7$
- What are HII, LINERS, Seyferts?

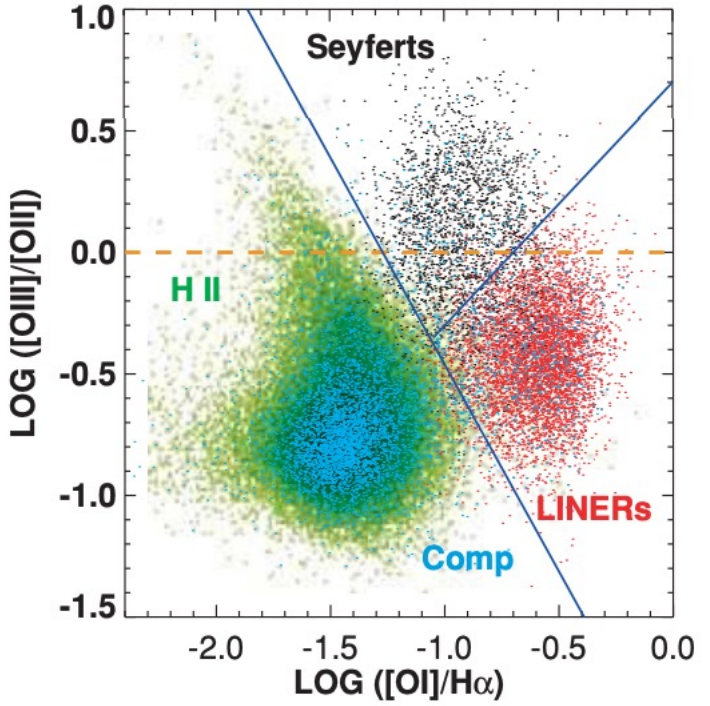


Figure 5. The [O III]/[O II] versus [O I]/Hα diagnostic diagram for SDSS galaxies with S/N > 3. Galaxies have been classified using the standard BPT diagnostic diagrams (Fig. 4). Ambiguous galaxies are not included. Our new preferred classification scheme is shown in blue. The H80 LINER line (orange dashed) is also shown.

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SDSS Navigate Visual Tool

- https://skyserver.sdss.org/dr18/VisualTools/navi
- Select an emission line galaxy with a measured spectrum
- Using the flux (EW), place the objects on the $\log\left(\frac{[OIII]}{[OII]}\right)$ vs $\log\left(\frac{[OI]}{H\alpha}\right)$ plot
  - Wavelengths of the emission lines: [O I] λ6300, [O II] λλ3726, [O III] λ5007, [Hα] λ6564
  - λλ indicates that you must sum across the doublet near those wavelengths
- Determine the type of emitter the galaxy is

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Reading a paper

- Order to read a paper:
  - Abstract
  - Figures
  - Conclusions
  - Introduction
  - The rest
- Identification of an [α/Fe]—Enhanced Thick Disk Component in an Edge-on Milky Way Analog
  - https://iopscience.iop.org/article/10.3847/2041-8213/abfc57
- Let’s start reading!

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Finding and storing papers

- ADS/NASA
  - https://ui.adsabs.harvard.edu/
- ArXiv
  - https://arxiv.org/
- Pathfinder
  - https://huggingface.co/spaces/kiyer/pathfinder
- Zotero
  - https://www.zotero.org/
- Let’s decide your pairs for the presentation!

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LaTeX and Overleaf

- preferable typesetting software in academic writing
- For your written report, you must use LaTeX in either the PASA or MNRAS journal style
  - Log into overleaf
  - New Project → Templates → View All
  - Search either PASA (Cambridge) or MNRAS
- UNSW provides free Pro accounts, so make sure to sign up using your university email

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How do I start using LaTeX

- https://www.overleaf.com/learn/latex/Tutorials
- Sections
  - **\begin{section}**
    \end{section}
- Lists (ordered and unordered)
  - **\begin{itemize}** or **\begin{enumerate}**
    \end{itemize} or **\end{enumerate}**
- Images (including captions)
  - **\begin{figure}**
    \end{figure}

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How do I start using LaTeX

- Tables
  - Use table generators (e.g. https://www.tablesgenerator.com/)
- Equations (including the Physics package)
  - https://www.overleaf.com/learn/latex/Mathematical_expressions
- Code
  - **\begin{lstlisting}[language=Python]**
    \end{lstlisting}
- References
  - Export from Zotero and upload

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Setting up your document

- Create a folder for your figures, tables, and sections
- Import all the packages you require, for example:
  - **\usepackage{graphicx}**
  - **\usepackage{amsmath}**
  - **\usepackage{listings}**
- Have a clear storyline
- Figure captions go below, and table captions go above
- All captions must include the takeaway message of the figure and include all details required for a reader's understanding

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